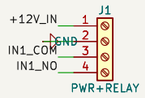
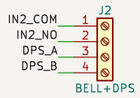
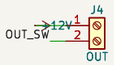
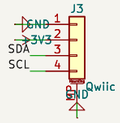
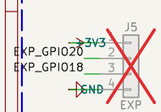
## CONNECTORS (field side)



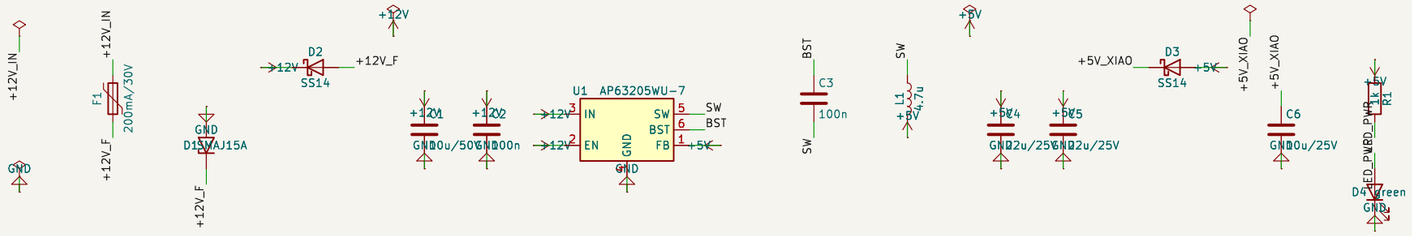




## POWER: 12 V input protection, 5 V buck, XIAO 5 V feed




Power budget: UA-Hub LINE output 12 V / 0.33 A max. F1 holds 200 mA.

D2 = reverse-polarity protection; D1 + F1 clamp surges / reversed input.

D3: Seeed requires a diode (anode = supply) when feeding the XIAO 5V pin.

Field loops: V_LOOP -> 2.2k -> opto LED -> COM ... contact ... NO -> 1k -> GND (~3.3 mA at 12 V, ~1.1 mA on USB).

Contact closed -> opto ON -> INx_SIG LOW (active low). RC 1k/10n + firmware debounce.

J2.3/J2.4 (DPS): JP1/JP2 1-2 = input (reed contact); 2-3 = SSR output to UA-Hub DPS input.

Never wire DPS input in parallel with a reed that already feeds the UA-Hub DPS input.

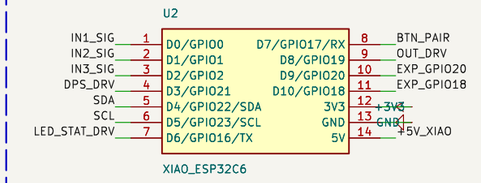
No mains voltage on this PCB. 230 V doorbells: use a certified coupling relay (electrician).

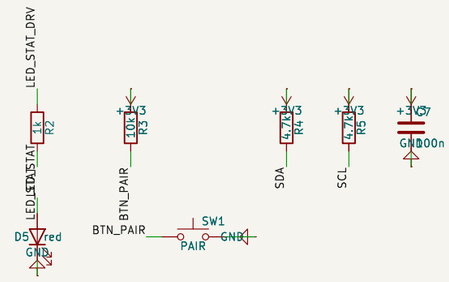
J6 USB-C (Rev B): 5 V power only, no data. Firmware upload uses the XIAO's own USB-C.

VBUS -> F2 -> D13 -> +5V_XIAO (OR with D3) and D16 -> V_LOOP (OR with D15 from +12V). J4 OUT+ is 12 V only.

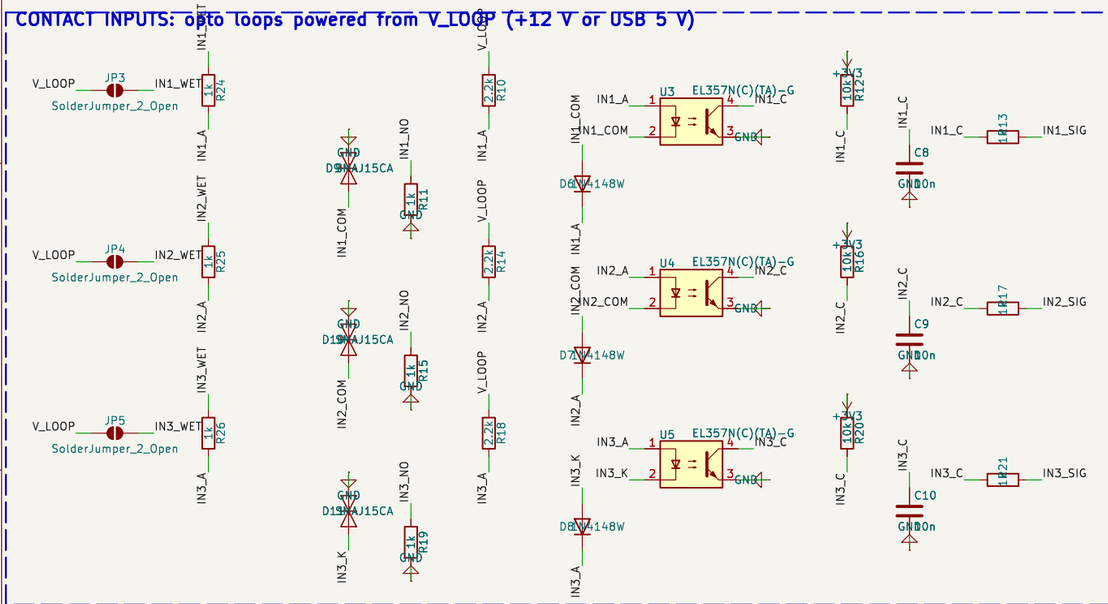
JP3-JP5 (open by default): bridge to add R24-R26 (1k) in parallel with R10/R14/R18, loop current ~6 mA for silver-contact relays.

## MCU: XIAO ESP32C6 (socketed)

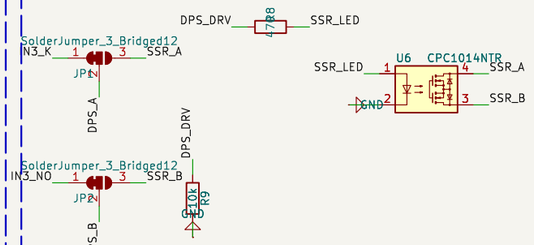


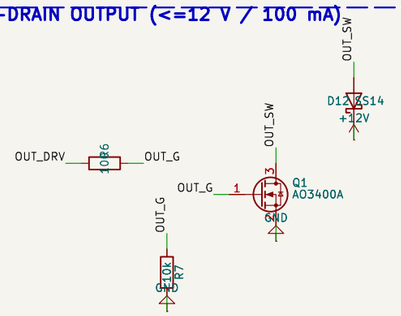

## CONTACT INPUTS: opto loops powered from V_LOOP (+12 V or USB 5 V)



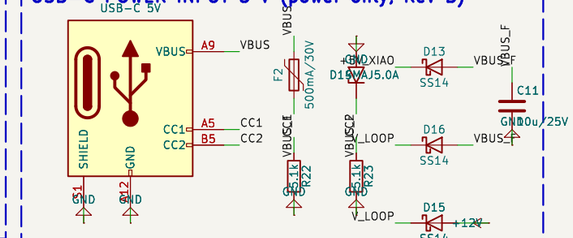

## DPS PORT: Input (default) or isolated SSR output




## OPEN-DRAIN OUTPUT (<=12 V / 100 mA)




## USB-C POWER INPUT 5 V (power only, Rev B)




License: CERN-OHL-S-2.0
Zigbee dry-contact bridge for UniFi Access UA-Hub, Seeed XIAO ESP32C6
**AccessBridge project**

Sheet: /
File: accessbridge.kicad_sch

***Title: AccessBridge C6***

| Size: A3 | Date: 2026-10-18 | **Rev: B (HW-1.1)** |
|---|---|---|
| KiCad E.D.A. 9.0.9 | | Id: 1/1 |